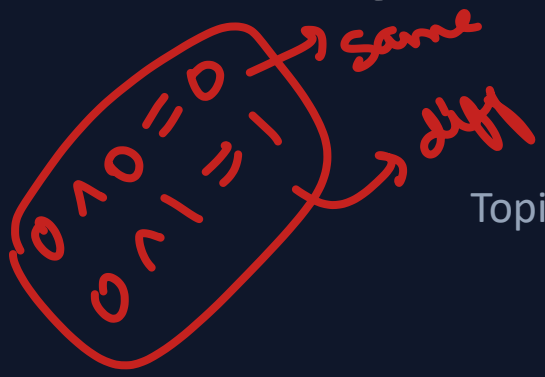
BASED ON LATEST OCTOBER PATTERN

Infosys Pseudo Code Questions



Topic: Bitwise XOR Logic & Operations

Question 01

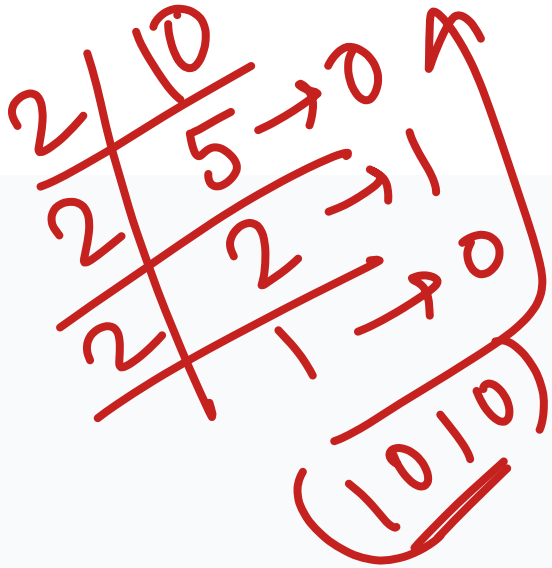
```
Set Integer a = 10
Set Integer b = 5
Set Integer result = a ^ b
display result
```

A) 5

B) 10

C) 15

D) 0



Answer: C) 15

```
Set Integer a = 10
Set Integer b = 5
Set Integer result = a ^ b
display result
```

XOR Rule: Returns 1 if bits differ.

`a` (10) is binary **1010**.

`b` (5) is binary **0101**.

Binary **1111** equals decimal 15.

```
  1 0 1 0 (a)
^ 0 1 0 1 (b)
  -------
  1 1 1 1 (15)
```

Question 02

```
Set Integer x = 12
Set Integer res = (x ^ x) ^ (x ^ 0)
display res
```

A) 0

B) 12

C) 24

D) Error

Answer: B) 12

Property 1

`x ^ x = 0`.

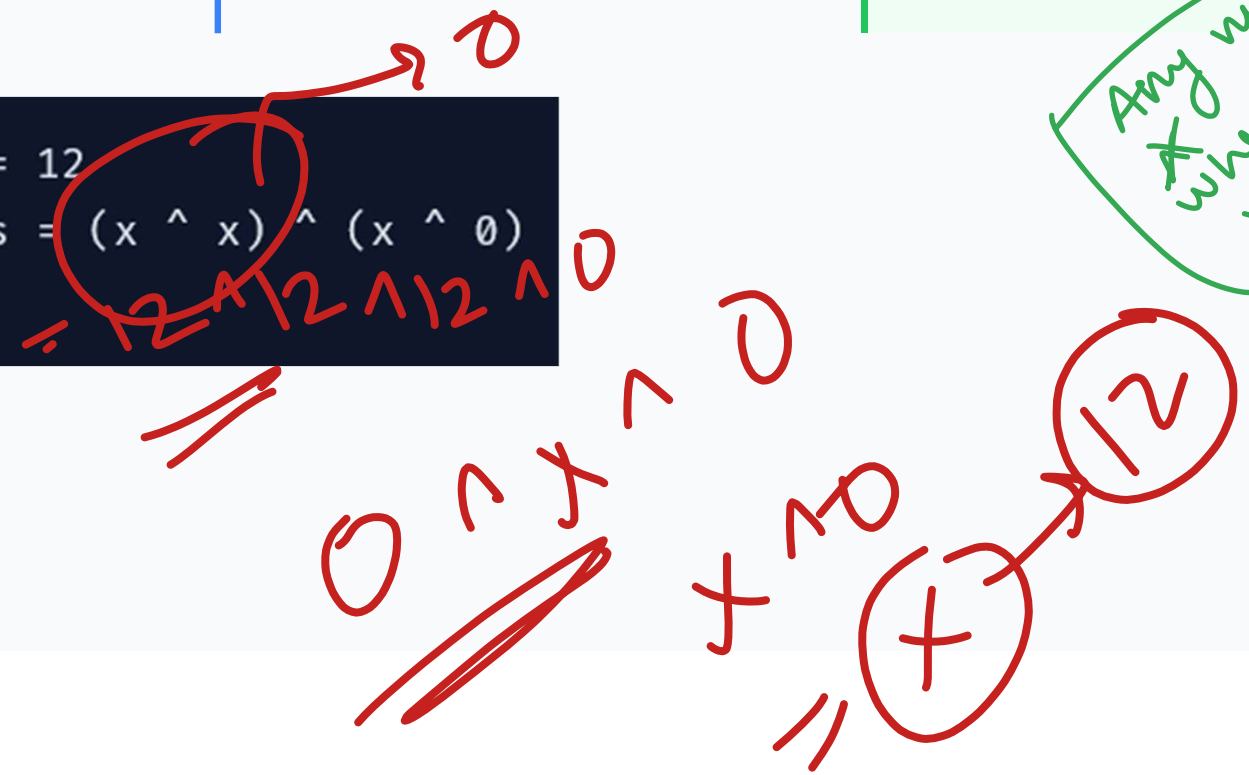
Property 2

`x ^ 0 = x`.

Resolution

`0 ^ 12 = 12`.

```
Set Integer x = 12
Set Integer res = (x ^ x) ^ (x ^ 0)
display res
```



Question 03

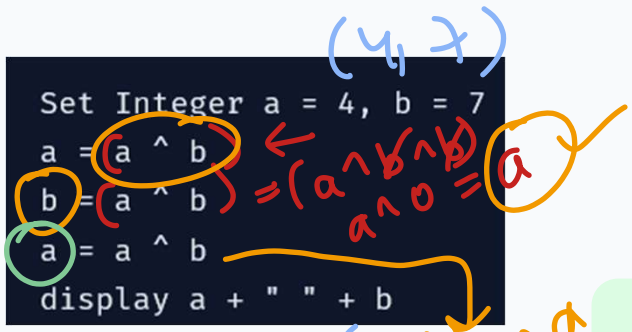
```
Set Integer a = 4, b = 7
a = a ^ b
b = a ^ b
a = a ^ b
display a + " " + b
```

A) 4 7

B) 7 4

C) 0 0

D) 3 3



Swapping 2 no.s without ^ temp variable.

Answer: B) 7 4

Classic XOR Swap Algorithm

1. `a = 4 ^ 7`
2. `b = (4 ^ 7) ^ 7` — b becomes 4
3. `a = (4 ^ 7) ^ 4` — a becomes 7
4. **Values swapped.**

Question 04

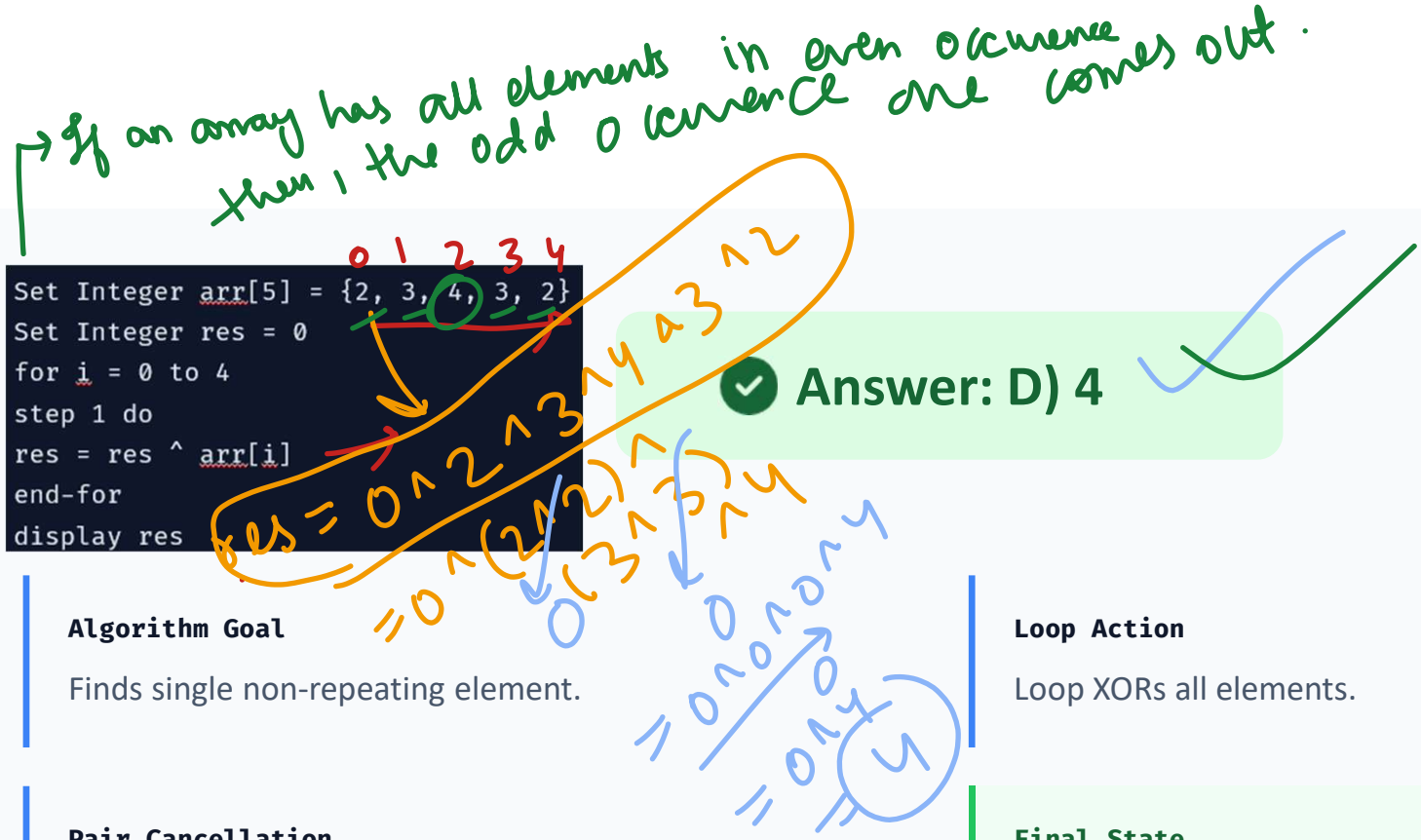
```
Set Integer arr[5] = {2, 3, 4, 3, 2}
Set Integer res = 0
for i = 0 to 4
step 1 do
res = res ^ arr[i]
end-for
display res
```

A) 0

B) 2

C) 3

D) 4



Answer: D) 4

Algorithm Goal

Finds single non-repeating element.

Loop Action

Loop XORs all elements.

Pair Cancellation

Pairs cancel out to 0.

Final State

Result is 4.

Question 05

```
Set Integer a = 3
Set Integer b = 6
Set Integer res = (a++) ^ (++b)
display res
```

A) 4

B) 5

C) 2

D) 0

```
Set Integer a = 3
Set Integer b = 6
Set Integer res = (a++) ^ (++b)
display res
```

Answer: A) 4

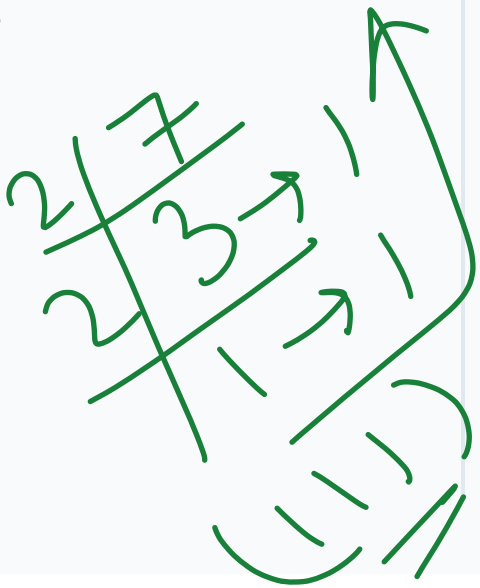
a++ uses initial value 3.

++b uses incremented value 7.

Expression evaluates: **3 ^ 7**.

Binary `011 ^ 111 = 100` (4).

```
  0 1 1 (3)
^ 1 1 1 (7)
  -----
  1 0 0 (4)
```



Question 06

```
Set Integer res = 0
for i = 1 to 3
step 1 do
res = res ^ i
end-for
display res
```

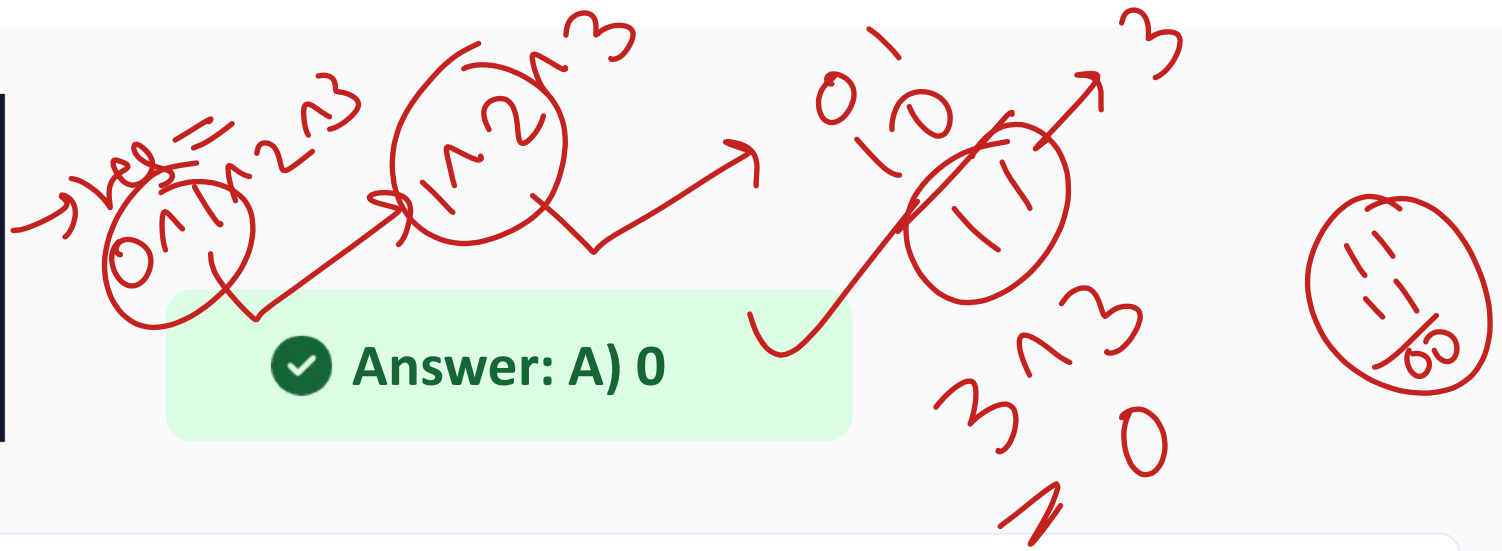
A) 0

B) 1

C) 2

D) 3

```
Set Integer res = 0
for i = 1 to 3
step 1 do
res = res ^ i
end-for
display res
```



Answer: A) 0

i = 1: `0 ^ 1 = 1`

i = 2: `1 ^ 2 = 3`

i = 3: `3 ^ 3 = 0`

Question 07

```
Set Integer x = 5, y = 3
Set Integer res = (x ^ y) + (x & y)
display res
```

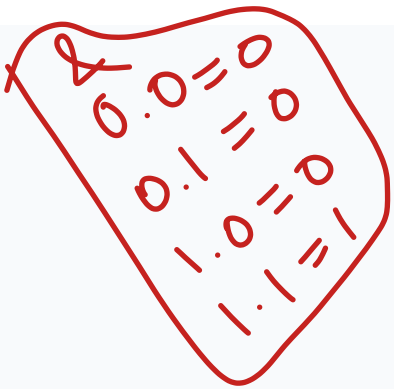
A) 8

B) 7

C) 6

D) 2

Answer: B) 7



```
Set Integer x = 5, y = 3
Set Integer res = (x ^ y) + (x & y)
display res
```



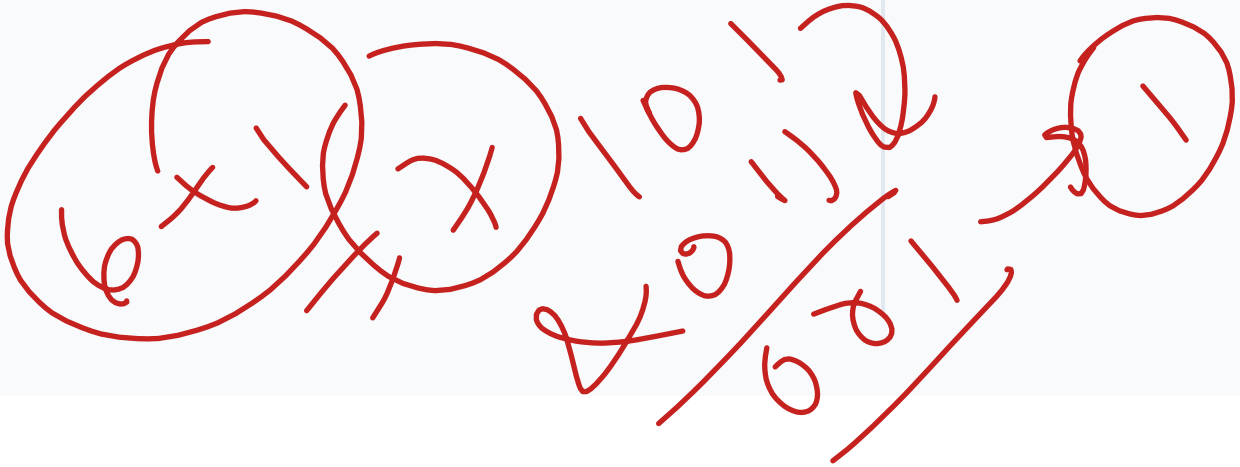
`x` (5) is **101**.

`y` (3) is **011**.

`x ^ y = 101 ^ 011 = 110 (6)`

`x & y = 101 & 011 = 001 (1)`

Total sum: 6 + 1 = 7.



Question 08

```
Set Integer a = 12
Set Integer b = 10
if ((a ^ b) == 0)
then do
display "Match"
else
do
display "Mismatch"
end-if
```

A) Match

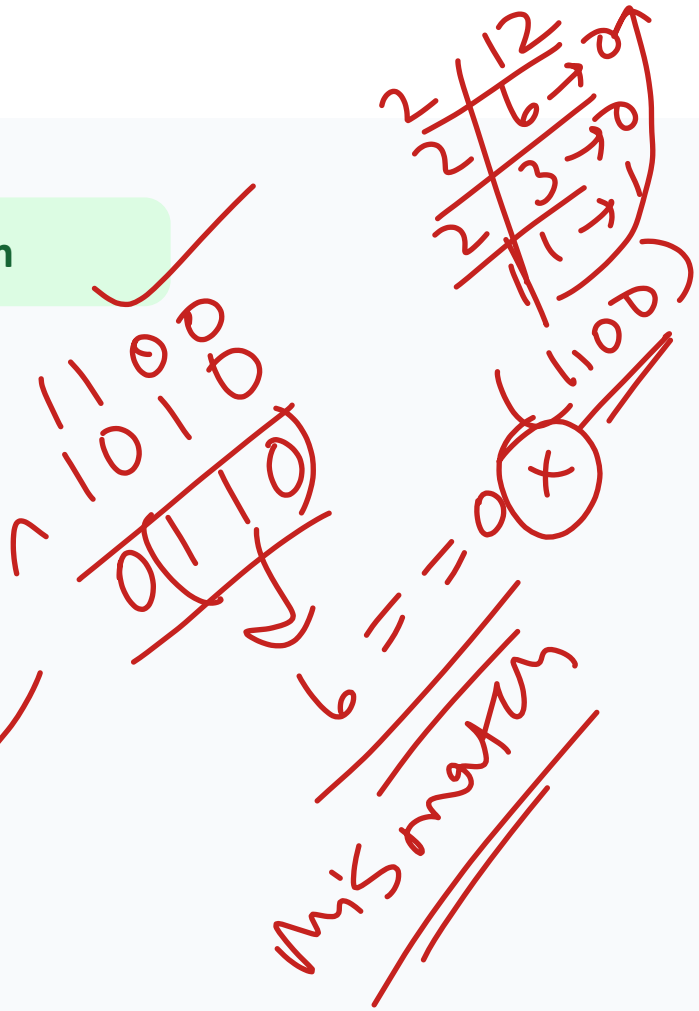
B) Mismatch

C) Error

D) None of the above

Answer: B) Mismatch

```
Set Integer a = 12
Set Integer b = 10
if ((a ^ b) == 0)
then do
display "Match"
else
do
display "Mismatch"
end-if
```



Equality Check

XOR checks for equality.

Bit Evaluation

`12 ^ 10 = 6`.

Branching

`6 == 0` is False. Else block prints "Mismatch".

Question 09

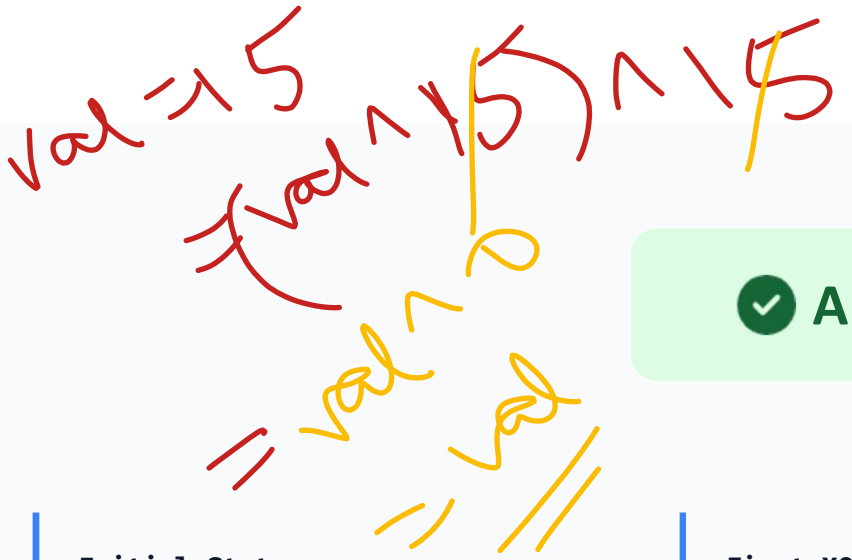
```
Set Integer val = 15
val = val ^ 15
val = val ^ 15
display val
```

A) 0

B) 15

C) 30

D) 1



Answer: B) 15

Initial State

Initial value: 15.

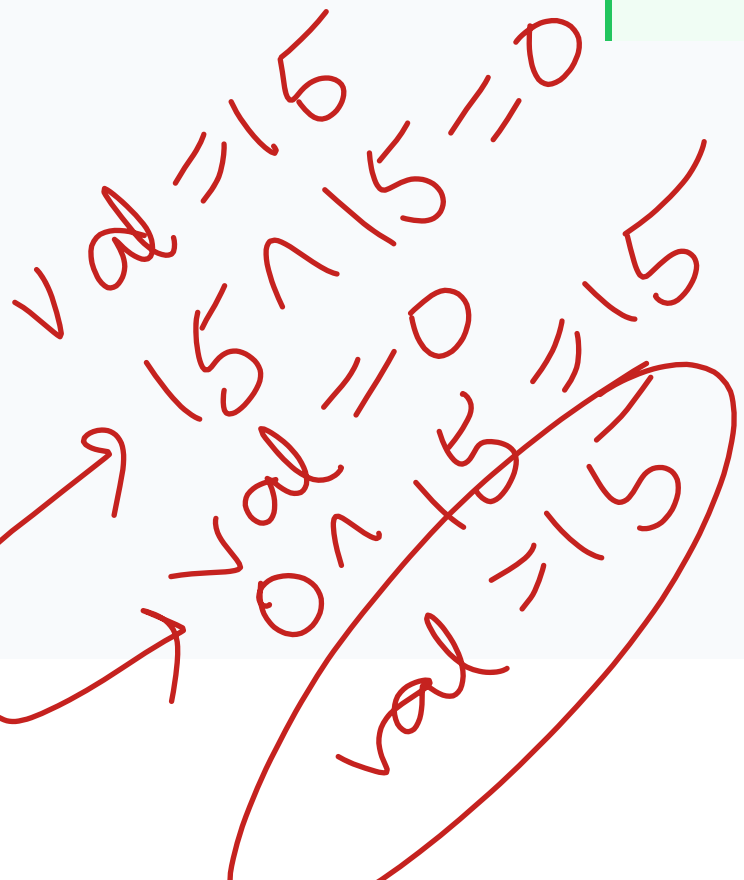
First XOR

`15 ^ 15 = 0`.

Second XOR

`0 ^ 15 = 15`.

```
Set Integer val = 15
val = val ^ 15
val = val ^ 15
display val
```



Question 10

```
Set Integer a = 6
Set Integer b = 2
Set Integer c = a ^ b
display c
```

A) 8

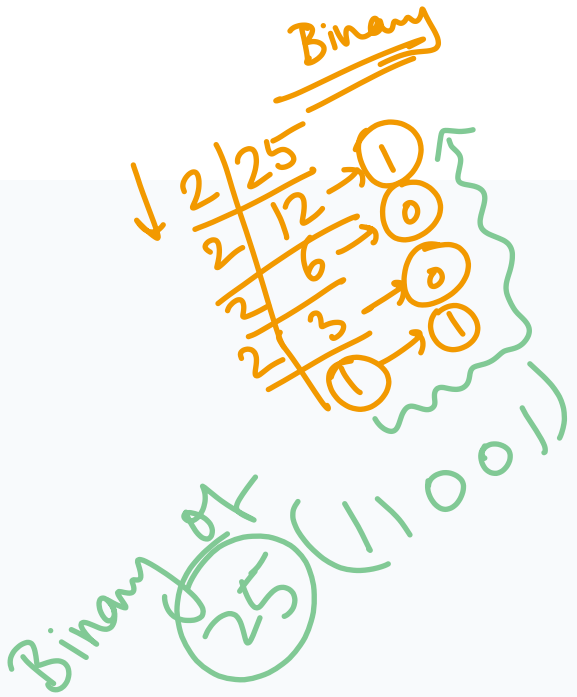
B) 6

C) 4

D) 2

```
Set Integer a = 6
Set Integer b = 2
Set Integer c = a ^ b
display c
```

Answer: C) 4





Binary Conversion

`a` (6) is binary **110**.

`b` (2) is binary **010**.

XOR Operation

`110 ^ 010 = 100`

Binary 100 is decimal 4.



End of Practice Set

Great job! Keep practicing for your technical rounds.